Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 0.1 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A phase Ib/IIa, randomized, double-blind, placebo-controlled, multiple-ascending dose, parallel-group study to investigate the safety, tolerability, pharmacokinetics, and pharmacodynamics of RO7126209 following intravenous infusion in patients with prodromal or mild to moderate alzheimer's disease **Short Title/Acronym:** Brainshuttle AD Study **Protocol Number:** BP42155 (2020-002477-98) **NCT Number:** NCT04639050 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** RO7126209 (trontinemab) **Phase of Development:** Phase I/II (Phase 1b/2a) **Medical Monitor:** Clinical Operations Lead / Therapeutic Area Leader Dementias, Hoffmann-La Roche

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, immunogenicity, pharmacokinetics, and pharmacodynamics of multiple-ascending intravenous (IV) doses of RO7126209 in participants with prodromal or mild to moderate Alzheimer's disease (AD). **Secondary Objectives:** To assess the efficacy of RO7126209 in reducing amyloid plaque burden (measured via PET) and to evaluate changes in downstream fluid biomarkers (e.g., CSF and plasma p-tau181, p-tau217, total tau, neurogranin, and Abeta42/40 ratio).

2.2 Background & Rationale Alzheimer's disease is driven by the buildup of beta-amyloid plaques in the brain. Trontinemab is an investigational bispecific 2+1 amyloid-beta antibody engineered to more easily cross the blood-brain barrier using Roche's "brain shuttle" technology. This study is being conducted to evaluate whether this enhanced brain access enables rapid and robust reduction of amyloid and improvement in downstream biomarkers compared to traditional therapies in patients with AD.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (Parallel-group)

3.2 Planned Enrollment & Duration Target N: >160 participants (Staggered cohorts of ~60 participants initially, with expansions up to >60 per dose cohort in Part 2) **Number of Sites:** Global, Multicenter **Estimated Study Start:** 15-Mar-2021 **Estimated Primary Completion:** 30-Jun-2030

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 50 to 85 years (inclusive). 2. Diagnosed with prodromal Alzheimer's disease, or mild to moderate Alzheimer's disease. 3. Confirmed amyloid positive based on an amyloid positron emission tomography (PET) scan.

4.2 Exclusion Criteria 1. Evidence of large areas of superficial siderosis or probable cerebral amyloid angiopathy (CAA) at MRI screening. 2. Any unstable medical disorder that can impact participant safety or study outcomes. 3. Concurrent or recent treatment with other investigational anti-amyloid therapies.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple ascending doses tested (e.g., 0.2 mg/kg, 0.6 mg/kg, 1.8 mg/kg, and 3.6 mg/kg) administered every 4 weeks. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** 28 weeks for the primary treatment period (followed by a 28-week safety follow-up period).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care medications for AD and pre-medications to manage potential infusion-related reactions. **Prohibited:** Other investigational drugs or biological products.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, MRI screening (for ARIA/CAA), Amyloid PET Scan
Baseline	Day 0	Randomization, First IV Infusion, Clinical Safety Assessments
Interim	Weeks 4 to 24	IV Infusions (every 4 weeks), MRI monitoring, CSF/Plasma biomarker extraction
End of Study	Week 28	Final Primary Amyloid PET scan, Transition to 28-week safety follow-up

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Assessment of safety, tolerability, PK, and PD, specifically evaluating the rapid reduction of amyloid plaque (mean change from baseline in centiloids) at 28 weeks. **Measurement Method:** Amyloid Positron Emission Tomography (PET) scanning and clinical monitoring.

7.2 Secondary & Exploratory Endpoints * Change from baseline in cerebrospinal fluid (CSF) and plasma biomarkers (Abeta42/40 ratio, p-tau181, p-tau217, total tau, and neurogranin). * Immunogenicity and long-term pharmacokinetic profiling.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine hospital visits to monitor infusion-related reactions and Amyloid-Related Imaging Abnormalities (ARIA-E for edema and ARIA-H for microhemorrhages/ hemosiderin deposition) via MRI. **Serious Adverse Events (SAEs):** Evaluated strictly, including severe ARIA complications or intracerebral macrohemorrhages; reporting required within standard 24-hour safety timelines. **Data Monitoring Committee (DMC):** External/Internal committees reviewing unblinded safety and PK/PD data to guide dose expansion in Part 2.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Set for efficacy/ biomarkers and Safety Set for AE/ARIA tracking. **Sample Size Justification:** Staggered, parallel-group design adapted to capture sufficient PK/PD profile across 4 initial cohorts, eventually expanding to >60 participants per cohort in Part 2 to ensure adequate safety profiling. **Handling of Missing Data:** Standard imputation methods for longitudinal biomarker trajectories.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite safety monitoring and recurrent MRI check-ups at designated hospital visits. **Audit Trail:** Centralized reading of PET/MRI scans and standardized biofluid analysis using the Roche NeuroToolKit.

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11. Study Identifiers & Governance

Primary ID: BP42155 **Registry Number:** NCT04639050 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** Brainshuttle AD **Official Regulatory Title:** A phase Ib/IIa, randomized, double-blind, placebo-controlled, multiple-ascending dose, parallel-group study to investigate the safety, tolerability, pharmacokinetics, and pharmacodynamics of RO7126209 following intravenous infusion in patients with prodromal or mild to moderate alzheimer's disease

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12. Clinical Framework (Alzheimer's Disease Specific)

Primary Condition: Prodromal or Mild to Moderate Alzheimer's Disease **Diagnosis Threshold:** Amyloid positive based on PET scan **Medical Methodology:** Brainshuttle bispecific 2+1 amyloid-beta monoclonal antibody **Optimization Target:** Amyloid plaque clearance and downstream neuro-biomarker reduction

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13. Study Procedures & Methodology

Management Pathway: Intravenous (IV) Infusion every 4 weeks **Education Protocol (HCP):** ARIA management, MRI monitoring procedures, and safety monitoring protocols **Education Protocol (Patient):** Informed consent detailing potential for infusion-related reactions, ARIA, and required hospital visits **Quality Audit Mechanism:** Centralized evaluation of amyloid PET centiloid levels and Roche NeuroToolKit CSF/plasma assays

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14. Observation & Follow-Up Schedule

Total Duration: 68 weeks for Parts 1 & 2 (up to 129 weeks if participating in extended cohorts) **Onsite Visit Cadence:** Every 4 weeks matching IV infusion schedule, extending into the 28-week safety follow-up **Remote Monitoring Frequency:** Between hospital visits as clinically indicated **Checklist Review Interval:** Regular clinical and neurological assessments conducted at each hospital visit

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] | ___ |** ***[DD-MMM-YYYY] | | Lead Statistician | [Name] |*** _____ | [DD-MMM-YYYY] |